Intelligent Decision Support System for Sri Lankan Tea Estate Management

COMMON INTEGRATED SOLUTION REPORT

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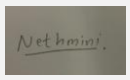
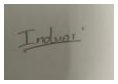
Sri Lanka

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ABSTRACT

Tea estate management in Sri Lanka requires timely decisions regarding workforce welfare, crop health, weather-related risks, and processed-tea quality. However, these activities are often managed separately using manual records and individual experience, limiting the availability of consistent information for decision-making. This report presents the Intelligent Decision Support System for Sri Lankan Tea Estate Management, a web-based platform that brings together four decision-support components: Labor Management and Welfare Optimization, Tea Leaf Disease Detection and Pattern Analysis, Tea-Specific Weather and Climate Intelligence, and AI-Based Tea Sorting and Grading Automation. The labor component uses a LightGBM classifier to classify worker conditions using heart rate and blood oxygen saturation data. The disease detection component applies YOLOv8s to tea leaf images to identify diseases and provide confidence scores and treatment recommendations. The weather component uses a Gradient Boosting Regressor for weather forecasting and a Multi-output Random Forest Regressor for disease-risk estimation. The tea grading component uses the ConvNeXtV2-Tiny model to classify processed-tea images into grade categories. The platform uses a React.js frontend, a Node.js/Express.js backend, MongoDB storage, and model-serving services. Component-level evaluation results reported in the project include 82% accuracy for the labor model and 96.1% mAP@0.5, 95.2% precision, 93.8% recall, and 94.5% F1-score for the disease detection model. The weather and grading results require interpretation alongside their corresponding evaluation evidence. The four components were tested together in the running application; formal integration-test records and user-validation evidence should be documented to support a complete assessment of system-level performance. Overall, the project demonstrates a multi-component approach to tea-estate decision support while identifying real-world validation, interface standardization, data security, and operational reliability as important areas for further improvement.

Keywords: decision support system, tea estate management, machine learning, deep learning, Internet of Things

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Finally, the authors express their appreciation to all members of group R26-IT-118 for their collaboration and contributions to the development of the Intelligent Decision Support System for Sri Lankan Tea Estate Management.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CI/CD	Continuous Integration and Continuous Delivery
DL	Deep Learning
DSS	Decision Support System
HR	Heart Rate
IoT	Internet of Things
LightGBM	Light Gradient Boosting Machine
mAP	Mean Average Precision
ML	Machine Learning
NFR	Non-Functional Requirement
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SpO_2	Blood Oxygen Saturation (peripheral)
UAT	User Acceptance Testing
UI	User Interface
YOLO	You Only Look Once (object-detection model family)

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

The tea industry is an important part of Sri Lanka's agricultural economy, contributing to export earnings, employment, and rural livelihoods. Ceylon tea is also internationally recognized for its quality [1]. However, tea estate management faces several challenges, including labour availability, tea leaf diseases, changing weather conditions, and the need to maintain consistent product quality. These challenges can affect productivity and increase production costs [1]. Previous research has also examined the vulnerability of Sri Lankan tea production to climate variability [4], highlighting the importance of reliable weather-related information in estate planning and management.

Decision-making in a tea estate involves several interconnected operational areas. Labour allocation requires information about workforce availability and worker welfare. Disease management requires the identification of affected tea leaves and appropriate treatment recommendations, while weather information can support the assessment of environmental conditions relevant to crop management. Similarly, processed-tea grading requires consistent assessment of product quality. When information for these activities is maintained separately or recorded manually, managers may face difficulties obtaining relevant information efficiently and making timely decisions.

Advances in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and the Internet of Things (IoT) provide opportunities to support agricultural decision-making through data-driven analysis and intelligent monitoring. Previous research has reviewed the application of deep learning in agriculture [2], while Mohanty et al. demonstrated the use of deep learning for image-based plant disease detection [3].

weather forecasting, and product classification [1], [2]. Deep learning techniques have also been applied to image-based plant disease detection [3]. Nevertheless, addressing these operational areas through separate tools may limit the availability of their outputs within a common management environment.

To address this challenge, the Intelligent Decision Support System for Sri Lankan Tea Estate Management proposes a web-based platform that brings together four specialized components: Labor Management and Welfare Optimization, Tea Leaf Disease Detection and Pattern Analysis, Tea-Specific Weather and Climate Intelligence, and AI-Based Tea Sorting and Grading Automation. The platform provides a common environment through which estate managers can access the outputs of these components. By presenting information from the four decision-support areas within one application, the proposed system aims to support more informed management decisions while establishing a foundation for further integration and future development.

1.2 Stakeholders and user needs

The primary user of the proposed platform is the tea estate manager, who is responsible for coordinating workforce activities, supporting worker welfare, responding to crop diseases, monitoring weather-related risks, and maintaining processed-tea quality. The platform also addresses the needs of secondary stakeholders, including field supervisors, field workers, quality-control staff, and factory personnel, depending on the functions of each component.

Table 1.1 presents the key stakeholders and their anticipated needs in relation to the four components of the Intelligent Decision Support System for Sri Lankan Tea Estate Management. These needs were identified from the project documentation and relevant literature. However, direct stakeholder requirements elicitation, such as interviews, questionnaires, or surveys, is not documented in the available project materials. Therefore, the needs presented in this section should be understood as documentation-based requirements rather than requirements validated through direct stakeholder consultation.

Table 1.1: Stakeholders and user needs addressed by the platform

Stakeholder	Role in the platform	Need addressed	Component
Estate manager (primary)	Main dashboard user; makes allocation, treatment and quality decisions	Timely, consolidated information for decision making	All
Field supervisor	Receives job suggestions and worker-condition alerts	Fair task allocation and early notice of workers who may need attention	1
Field worker	Subject of wearable monitoring	Welfare protection; privacy and appropriate use of health-related data	1
Estate staff / plant-protection adviser	Uploads leaf images; reviews advice	Fast identification of leaf disease with treatment guidance that an expert can review	2
Estate manager planning field work	Reads forecasts and recommendations	Advance warning of weather-related disease risk	3
Quality-control / factory personnel	Uploads processed-tea images	Consistent, repeatable grade assessment	4

1.3 Alignment with the Sustainable Development Goals

The Intelligent Decision Support System for Sri Lankan Tea Estate Management has potential relevance to selected United Nations Sustainable Development Goals (SDGs) [7]. The proposed system aims to support workforce welfare, improve access to information for agricultural decision-making, and assist in monitoring crop health, weather-related risks, and processed-tea quality. These objectives may contribute to aspects of sustainable agricultural practices, responsible production, and worker well-being.

However, this alignment represents the project's intended contribution rather than a measured development impact. The system has not been evaluated through a field deployment or a formal impact assessment; therefore, its contribution to the selected

SDGs cannot yet be quantified. Table 1.2 presents the relevant SDGs and the potential contribution of the project. Section 7.5 further discusses possible negative effects, implementation considerations, and the conditions required to sustain the anticipated benefits.

Table 1.2: Candidate SDG alignment of the platform

SDG	Intended contribution (potential)	Component
SDG 3 – Good Health and Well-being	SDG 3 – Good Health and Well-being	SDG 3 – Good Health and Well-being
Supporting worker-welfare monitoring through physiological data analysis	Supporting worker-welfare monitoring through physiological data analysis	Supporting worker-welfare monitoring through physiological data analysis
Labor Management and Welfare Optimization	Labor Management and Welfare Optimization	Labor Management and Welfare Optimization
SDG 8 – Decent Work and Economic Growth	SDG 8 – Decent Work and Economic Growth	SDG 8 – Decent Work and Economic Growth
Supporting workforce management and worker welfare in tea-estate operations	Supporting workforce management and worker welfare in tea-estate operations	Supporting workforce management and worker welfare in tea-estate operations

1.4 Purpose and scope of the common report

This common report presents the integrated problem and proposed solution developed by research group R26-IT-118. The four individual theses provide detailed accounts of the research, model development, and experimental evaluation undertaken for each component. Rather than duplicating those component-level details, this report focuses on the overall system architecture, the relationships between the four components, their presentation within a common web-based platform, the available evaluation evidence, and the aspects of the integrated system that require further verification.

The components are numbered in this report according to the order used in the project paper [1]: Component 1, Labor Management and Welfare Optimization (Gunasekara L.M.N.P, IT22267986); Component 2, Tea Leaf Disease Detection and Pattern Analysis (Miuranga W.A.R, IT22209320); Component 3, Tea-Specific Weather and Climate Intelligence (Gunasekara G.N.D, IT22279620); and Component 4, AI-Based Tea Sorting and Grading Automation (Pathirana I.M, IT22282668). The order of the authors on the title page follows the declaration table and may therefore differ from the component numbering used throughout this report.

The report draws on the shared project paper [1] and the component summaries prepared by the group. Where differences exist between the available descriptions of a component, those differences are identified rather than combining inconsistent information into a single claim. The four components were tested together in the running application. However, the availability and completeness of formal integration-test records, interface specifications, and user-validation evidence must be considered separately when assessing the overall system. Any details that cannot be established from the available documentation are identified for confirmation rather than presented as verified findings.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

Operational decision-making in Sri Lankan tea estates involves four major areas: workforce allocation and welfare, tea crop health, weather conditions and their potential relationship with disease risk, and the quality of processed tea. These areas require different types of information and are often managed through separate activities and records. Consequently, estate managers may face difficulties accessing timely and consistent information to support evidence-based decisions across these operational areas [1].

The overall problem addressed by this project is the limited availability of a common, data-driven decision-support environment that brings information from these four areas together in one platform. To address this problem, the Intelligent Decision Support System for Sri Lankan Tea Estate Management combines four specialized components within a common web-based application. The proposed platform aims to improve access to component-level results and support informed operational decisions, while recognizing that further work may be required to establish direct data exchange and deeper functional integration between the components.

2.2 Root causes and overall research gap

Several related challenges contribute to the overall problem identified by the project. According to the project paper, many tea-estate operations rely on manual record-keeping and traditional management practices, which can make information gathering and decision-making time-consuming [1]. Workforce-related challenges, including labour shortages and uneven workloads, can further affect daily estate operations. The Tea Research Institute of Sri Lanka has also published guidance on labour welfare and productivity in tea estates [9]. Within the scope of this project, the labour-management component addresses worker-condition classification using physiological data, extending the system's focus beyond workforce allocation alone [1].

Tea leaf diseases present another challenge because they can affect crop health, yield, and quality. The project paper identifies a need to support disease identification alongside other tea-estate management activities [1]. Similarly, weather conditions, including temperature, rainfall, and humidity, are relevant to tea cultivation and may influence disease risk and estate-planning decisions. The project's weather-intelligence component aims to provide forecasts and information about potential disease-related risks to support management decisions [1]. In processed-tea production, traditional grading practices may depend on human inspection, while image-based classification offers an opportunity to support more consistent assessment. The project therefore includes an AI-based tea sorting and grading component as part of the wider decision-support platform [1].

The overarching research gap addressed by this project is the limited integration of these four decision-support areas within a common platform for tea-estate management. Rather than claiming that no integrated system exists, this report presents the gap in relation to the literature reviewed in the project paper and the four individual theses. The proposed system brings the four specialised components together within a shared web-based application, while further validation is required to establish its effectiveness in real-world tea-estate operations.

2.3 Subproblem and component mapping

The overall problem is divided into four distinct subproblems, each addressed by a dedicated component, as summarised in Table 2.1. Although the components focus on different operational areas, their outputs are relevant to the decisions made by tea-estate managers. Workforce condition, crop disease, weather-related risks, and processed-tea quality can all influence estate operations and management planning.

According to the project documentation, each component processes its own inputs and presents its outputs through the common web-based platform [1]. The four components were tested together in the running application. However, their current relationship is primarily established through the shared platform and the estate manager's use of the outputs, rather than direct data exchange between components.

Potential cross-component decision logic and deeper functional integration are considered areas for future development and are discussed in Section 8.3.

Table 2.1 maps each component to its corresponding subproblem, main inputs, outputs, and available evaluation evidence. The reported evaluation results should be interpreted according to the metrics and evaluation procedures used for each component, rather than as directly comparable measures of performance.

Table 2.1: Subproblem and component mapping

Component (owner)	Subproblem	Main input	Main output	Evidence available
1. Labor Management and Welfare Optimization - Gunasekara L.M.N.P (IT22267986)	Supporting workforce allocation and identifying worker-condition states during work	Worker information; heart rate (HR) and blood oxygen saturation (SpO_2) data	Worker-condition classification, alerts, and task-related suggestions	82% model accuracy across four worker-condition classes [1]
2. Tea Leaf Disease Detection and Pattern Analysis - Miuranga W.A.R (IT22209320)	Identifying tea leaf diseases and supporting appropriate treatment decisions	Tea leaf image	Predicted disease, confidence score, and treatment recommendations	mAP@0.5: 96.1%; precision: 95.2%; recall: 93.8%; F1-score: 94.5% [1]
3. Tea-Specific Weather and Climate Intelligence - Gunasekara G.N.D (IT22279620)	Providing weather forecasts and information about potential disease-related risks for tea-growing regions	Weather records and relevant weather readings	Weather forecasts, disease-risk estimates, and recommendations	An overall result of 89% is reported in the project documentation [1]. The metric's definition and evaluation procedure require clarification.

Component (owner)	Subproblem	Main input	Main output	Evidence available
4. AI-Based Tea Sorting and Grading Automation — Pathirana I.M (IT22282668)	Supporting consistent classification of processed tea into grade categories	Processed-tea image	Predicted grade and confidence score	Average precision, recall, and F1-score of 99.7% are reported [1]. The evaluation protocol and supporting results should be documented.

2.4 Overall aim and group objectives

Aim. To design and develop a web-based decision-support platform that supports Sri Lankan tea-estate managers in making timely and informed decisions regarding workforce welfare, tea leaf diseases, weather-related risks, and processed-tea quality.

The overall aim is addressed through the group objectives presented in Table 2.2. Each objective is mapped to the relevant component or components and to the evidence used to assess its achievement. The evaluation considers the system architecture, component-level model performance, application functionality, and integrated workflow. The reported metrics are interpreted according to the specific task and evaluation method of each component.

Table 2.2: Group objectives, contributing components and evaluation evidence

ID	Objective	Components	Evidence used for evaluation
O1	Design and implement a common web-based platform through which the four components present their outputs to the estate manager.	All four components	Architecture descriptions and diagrams (Chapter 3); implementation and integration evidence (Chapter 4).
O2	Classify worker conditions into Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress using heart rate (HR) and blood oxygen saturation (SpO_2) data, and support workforce-related decisions.	Component 1	Reported model accuracy; interface evidence for worker data and classification outputs.

ID	Objective	Components	Evidence used for evaluation
O3	Detect tea leaf diseases from uploaded images and provide the predicted disease name, confidence score, and treatment recommendations.	Component 2	Reported mAP@0.5, precision, recall, and F1-score.
O4	Forecast weather conditions and estimate disease-related risks for tea-growing regions, presenting relevant information to support estate management decisions.	Component 3	Reported forecasting and disease-risk evaluation results; model configuration and metric definitions.
O5	Classify processed-tea images into grade categories and present the predicted grade with a confidence score.	Component 4	Reported precision, recall, and F1-score; documented grade categories and model-service evidence.
O6	Evaluate the individual components using task-appropriate metrics and assess the integrated workflow through testing of the four components within the running application.	All four components	Component-level evaluation results (Chapter 6); available integration-testing evidence and documented test outcomes.
O7	Identify the validation, security, privacy, and operational requirements that should be addressed before deployment in real tea-estate environments.	All four components	Relevant discussions and evidence in Sections 6.4, 6.6, and 7.4.

2.5 Scope, assumptions and constraints

Scope

The proposed solution is a web-based decision-support platform designed for tea-estate management in Sri Lanka. It comprises four components: Labor Management and Welfare Optimization, Tea Leaf Disease Detection and Pattern Analysis, Tea-Specific Weather and Climate Intelligence, and AI-Based Tea Sorting and Grading Automation. The platform presents the outputs of these components within a common application to support estate managers in making informed operational decisions. It is intended to assist human decision-makers rather than replace their judgement and responsibility.

Assumptions

The operation of the platform assumes reliable internet connectivity between the web application, backend services, and externally hosted model services. The labor-management component assumes the availability of wearable devices or corresponding data sources capable of providing heart rate (HR) and blood oxygen saturation (SpO_2) data. The disease detection and tea grading components assume that uploaded images are sufficiently clear and suitable for classification. The weather-intelligence component assumes that relevant weather data are available for the regions being considered. The quality and availability of these inputs may influence the outputs produced by the system.

Exclusions

The platform is not intended to function as a medical diagnostic system. Worker-condition classifications are provided for decision-support purposes and require appropriate human interpretation. Similarly, disease-treatment recommendations are intended to support agricultural decision-making and do not replace advice from qualified agricultural professionals. A dedicated mobile application and additional IoT device capabilities are outside the current scope and are identified as potential future developments [1]. Physical tea-sorting machinery and automated hardware-based grading are also outside the scope of the solution described in the available project documentation.

Constraints

Model performance depends on dataset quality and evaluation methods, and results may differ under real tea-estate conditions. Dependence on third-party services may affect system availability. The limited project duration also restricted field deployment and real-world validation.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The proposed solution is a web-based decision-support platform designed for Sri Lankan tea-estate management. Through the web application, the estate manager can access the outputs of four intelligent components using relevant inputs, including worker information, leaf images, processed-tea images, and weather-related data. The platform brings together decision-support functions for workforce welfare, tea leaf disease detection, weather-related risk assessment, and processed-tea grading within a common application [1].

The main objective of this structure is to provide estate managers with a single point of access to information generated by the four components rather than requiring them to use separate applications for each decision area. Each component processes its relevant inputs and presents its outputs through the shared platform. Although the components were tested together in the running application, their current relationship is primarily based on the common platform and its users rather than direct data exchange between modules. Table 3.1 summaries the four components, their inputs, technologies, outputs, and reported evaluation evidence.

Table 3.1: Four-component overview

Component and owner	Main inputs	Model technology	Main outputs	Reported evaluation evidence
1. Labor Management and Welfare Optimization Gunasekara L.M.N.P (IT22267986)	Worker information, heart rate (HR), and blood oxygen saturation (SpO_2) data	LightGBM classifier with four worker-condition classes	Worker-condition classification, alerts, and task-related suggestions	82% accuracy [1]
2. Tea Leaf Disease Detection and Pattern Analysis Miuranga	Tea leaf images	YOLOv8s; hosted model inference through the implemented service	Predicted disease, confidence score, and treatment recommendations	mAP@0.5: 96.1%; precision: 95.2%; recall: 93.8%; F1-score: 94.5%

W.A.R (IT22209320)				[1]
3. Tea-Specific Weather and Climate Intelligence Gunasekara G.N.D (IT22279620)	Weather records and readings, including temperature, humidity, and rainfall	Gradient Boosting Regressor for weather forecasting and Multi-output Random Forest Regressor for disease-risk estimation	Weather forecasts, disease-risk estimates, and recommendations	An 89% result is reported [1]; the metric definition and evaluation procedure require clarification.
4. AI-Based Tea Sorting and Grading Automation Pathirana I.M (IT22282668)	Processed-tea images	ConvNeXtV2-Tiny deep-learning classifier	Predicted tea grade and confidence score	Average precision, recall, and F1-score of 99.7% are reported [1]; supporting evaluation evidence should be documented.

3.2 High-level architecture

The proposed platform follows a client–server architecture [1]. Users interact with a React.js web application, which sends requests to a Node.js backend implemented using Express.js. The backend processes application requests and communicates with the database and the relevant model-serving services to obtain and return results to the user interface [1].

Figure 3.1 illustrates the high-level architecture of the platform, including the web application, backend, data storage, and four intelligent components. The figure distinguishes the interactions described in the project documentation from architectural details that require further confirmation. In particular, it does not assume that all four components operate within a single runtime environment, use the same API structure, or share an identical database schema. The architecture therefore represents the common platform and the documented interactions without implying unverified implementation details.

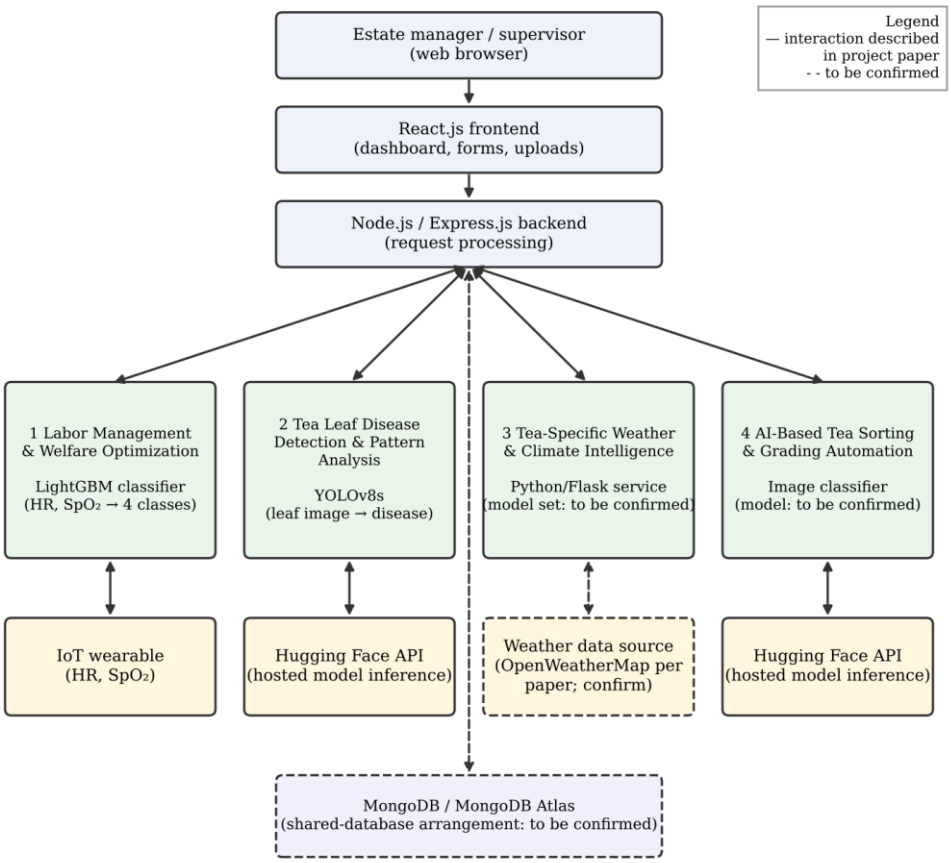


Figure 3.1: High-Level Architecture of the Integrated Decision Support Platform

3.3 Component 1 summary: Labor Management and Welfare Optimization

Problem. Daily tasks must be allocated fairly among workers, while worker conditions during field activities require appropriate monitoring. **Intended users.** Estate managers and field supervisors. **Inputs and data sources**. Worker details entered by the manager, including experience, skills, previous work, and availability, together with heart rate (HR) and blood oxygen saturation (SpO_2) data collected through wearable IoT devices [1]. **Algorithms and services**. Worker details are used to generate task-assignment suggestions. In addition, a LightGBM classifier (LGBMClassifier), trained using the PhysioNet Non-EEG Dataset for Assessment of Neurological Status in Google Colab, classifies physiological data into four categories: Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress [1]. **Outputs**. The predicted worker-condition class is displayed on the dashboard. The component also provides alerts when HR or SpO_2 values exceed configured thresholds and presents task-assignment suggestions [1]. **Interface and storage.** The

component uses a React.js frontend, a Node.js/Express.js backend, and MongoDB. According to the project documentation, device data are received by the Node.js server, checked, and stored [1]. **Evaluation evidence**. The project reports a classification accuracy of 82% [1]. **Limitations**. The classifier was trained using a public physiological dataset rather than data collected directly from tea-estate workers. Field validation involving estate workers is not documented in the available project materials. Furthermore, the four output categories represent stress-related classes and should not be interpreted as operational risk levels. **Contribution**. This component provides workforce-management and worker-welfare information within the common decision-support platform.

3.4 Component 2 summary: Tea Leaf Disease Detection and Pattern Analysis

Problem. Tea leaf diseases can affect crop yield and quality, making timely disease identification important for tea-estate management. **Intended users.** Estate managers and staff responsible for crop inspection. **Input**. A tea leaf image uploaded through the web application [1]. **Algorithm and service**. The uploaded image is transmitted from the React.js frontend to the Node.js backend and then submitted to a Hugging Face API hosting a YOLOv8s model. According to the project documentation, the system first checks whether the uploaded image contains a tea leaf. If the image is accepted, the model identifies the disease and assigns it to the corresponding category [1]. **Output**. The component returns the predicted disease name, confidence score, and treatment advice. These results are returned to the backend, stored in MongoDB, and displayed on the dashboard [1]. **Evaluation evidence.** The project reports a mean Average Precision at an Intersection over Union threshold of 0.5 (mAP@0.5) of 96.1%, precision of 95.2%, recall of 93.8%, and an F1-score of 94.5% from validation testing [1]. These metrics represent the disease-detection model's reported performance on its evaluation data and should not be interpreted as performance measures for the complete platform. **Limitations**. Model performance may vary depending on the dataset, image quality, disease categories represented, and field conditions. Treatment advice is intended to support decision-making and does not replace guidance from a qualified agricultural expert. The project paper does not describe the pattern-analysis functionality in detail; further information

should therefore be drawn from the individual component thesis. **Contribution**. This component provides crop-health information through tea leaf disease detection and associated treatment recommendations within the common decision-support platform.

3.5 Component 3 summary: Tea-Specific Weather and Climate Intelligence

Problem. Weather conditions influence tea growth and disease risk, creating a need for region-specific forecasts and information to support estate management decisions. **Intended users**. Estate managers responsible for planning field operations. **Inputs and data sources**. The project documentation identifies weather data for tea-growing regions, including Nuwara Eliya, Kandy, and Ratnapura. Temperature, humidity, and rainfall are listed in one section, while wind speed and an OpenWeatherMap data source are mentioned elsewhere [1]. **Algorithms and services**. The project paper describes different model configurations, as summarised in Table 3.2. The intended configuration for this component is a Gradient Boosting Regressor for weather forecasting and a Multi-output Random Forest Regressor for disease-risk estimation. The final description should be consistent with the implementation and evaluation evidence. **Outputs**. The component provides weather forecasts, disease-risk estimates, and recommendations intended to support estate-management decisions, including irrigation planning, fertilizer use, harvesting, and disease prevention [1]. **Evaluation evidence**. The project paper reports an overall prediction result of 89% [1]. The precise metric represented by this figure should be established from the component's evaluation results before it is described as accuracy. **Limitations**. The dataset details, regional coverage, model configuration, and definition of the reported 89% result should be verified against the individual component thesis and implementation. Forecasting performance may also depend on the quality and availability of weather data and the period represented in the training dataset. **Contribution**. This component provides weather forecasts and weather-related disease-risk information to support forward-looking decisions within the common tea-estate management platform.

3.6 Component 4 summary: AI-Based Tea Sorting and Grading Automation

Problem. Manual tea grading relies on experienced personnel and may produce inconsistent classifications. **Intended users**. Quality-control staff, factory personnel, and estate managers. **Input**. An image of processed tea uploaded through the web application [1]. **Algorithm and service**. The uploaded image is transmitted from the React.js frontend to the Node.js backend, which forwards it to a Hugging Face API hosting a deep-learning classification model. The predicted grade and confidence score are returned to the backend and stored in MongoDB [1]. The project paper identifies the model as ConvNeXtV2-Tiny (convnextv2_tiny.fcmae_ft_in22k_in1k) [8], trained on a processed-tea grade image dataset. **Output**. The dashboard presents the uploaded image, predicted tea grade, and confidence score [1]. **Evaluation evidence**. The project paper reports average precision, recall, and F1-score values of 99.7% [1]. These results should be interpreted in the context of the evaluation protocol and supporting experimental evidence. **Limitations**. Three issues require further verification. First, Table IV of the project paper lists eight grade classes (BM, BOP, BP, BROKEN_TEA, DUST, FANNING_2, PF, and PW_DUST), whereas the component documentation lists nine classes (BM, BOP, BT, BP, Broken Tea, Dust, Fanning 2, PF, and PW Dust). The final grade list should be reconciled against the actual model configuration and application. Second, the component documentation describes integration with a remotely hosted classification service, while the original training scripts, complete dataset documentation, and model weights were not available in the supplied source code. Therefore, the available evidence supports the integration of the hosted service more clearly than it establishes the complete model-training process. Third, the reported high evaluation scores require examination of the evaluation protocol, including the separation of training and test images and the distribution of samples across classes. **Contribution**. This component provides processed-tea quality classification to support quality-related decision-making within the common platform.

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Table 3.3: Comparison of the four components

Aspect	Labor and welfare	Disease detection	Weather and climate	Sorting and grading
Task type	Tabular classification (4 classes)	Object detection in images	Forecasting and risk prediction	Image classification (grades)
Input data	HR, SpO_2; worker details	Leaf image	Weather data	Processed-tea image
Inference location	Platform backend (to be confirmed)	Hugging Face API	Python service (to be confirmed)	Hugging Face API
Storage	MongoDB	MongoDB	To be confirmed	MongoDB / MongoDB Atlas
Reported metric	Accuracy 82%	mAP@0.5, P, R, F1	Accuracy 89%	Average P, R, F1
Main caveat	Public dataset; not estate workers	Dataset and field conditions	Model configuration unconfirmed	Grade list and training provenance unconfirmed

3.7 End-to-end user and data flow

Figure 3.2 illustrates the movement of user inputs and data through the platform. In each component, the user provides or initiates the relevant input, the backend forwards the request or data to the corresponding service, and the resulting output is returned for storage and display on the dashboard, where applicable.

For Component 1, inputs include worker and task details entered by the estate manager and heart rate (HR) and blood oxygen saturation (SpO_2) data received from wearable devices. Component 2 processes an uploaded tea leaf image, which is checked for the presence of a tea leaf before disease detection is performed. Component 3 uses regional weather data or requests to generate weather forecasts and disease-risk information. Component 4 processes an uploaded image of processed tea to predict its grade.

The four workflows operate as separate component-level processes within the common web application, and their outputs are made accessible through the dashboard. The available project documentation does not describe a workflow in

which the output of one component is directly used as the input to another. Therefore, the current integration is represented as a shared application and user interface rather than direct data exchange between the four components.

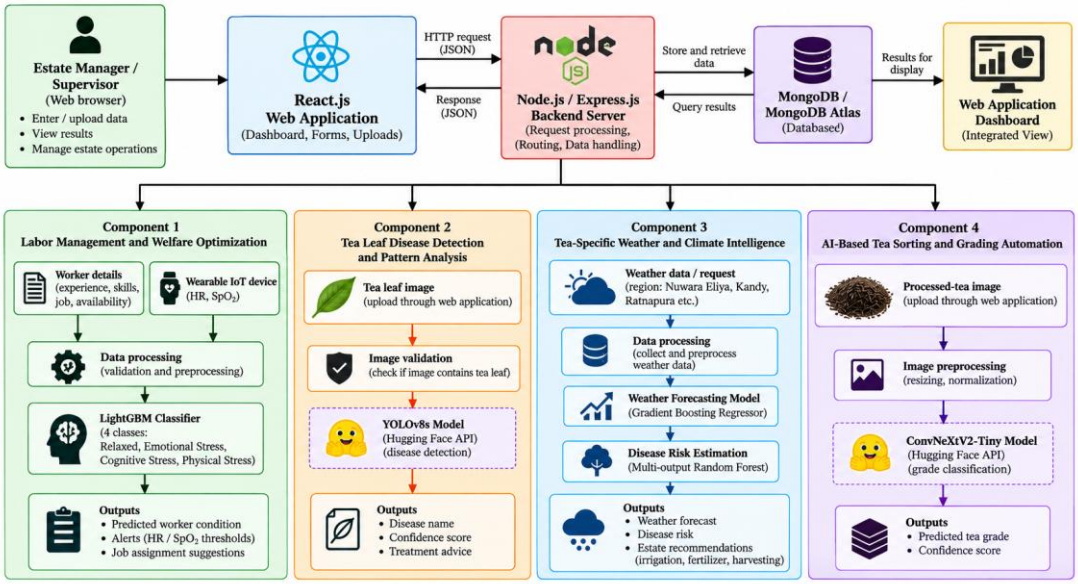


Figure 3.2: End-to-End Data and Decision Workflow of the Platform

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The project paper describes a modular, loosely coupled architecture in which each component processes its own data and presents its results through the common dashboard [1]. Integration is therefore primarily achieved through the shared presentation layer and backend request handling. The React.js application provides a common entry point for users, while the Node.js/Express.js backend handles application requests and communicates with the database, external APIs, and the relevant intelligent services [1].

Table 4.1 summaries the interfaces on which this integration approach depends. The available project documentation does not fully specify the integration sequence, deployment environments, interface ownership, or procedures used to manage integration risks. These details should be documented where supporting evidence is available.

As a recommendation for further development, an interface-first approach could improve consistency between components implemented using different technologies, including Node.js, Python, and hosted model-inference services. Under this approach, the producer and consumer of each interface would agree on the request route, payload structure, response format, and error-handling conventions before implementation. This would help reduce interface mismatches and simplify integration testing across the four components.

4.2 Interfaces, data contracts and shared models

Table 4.1 summaries the interfaces identified from the architecture and component descriptions in the project documentation. It records the producer and consumer of each interface, the associated inputs and outputs, the storage arrangements described, and whether a formal interface contract is documented in the available evidence. A formal contract would normally specify the endpoint or communication method, payload structure, authentication requirements, response format, and error-handling conventions.

The available project documentation does not provide complete formal interface contracts for the interfaces listed in Table 4.1. The team should document the relevant details in Appendix A where implementation evidence is available. The absence of documented contracts should not be interpreted as proof that the interfaces are not implemented.

Table 4.1: Component interface and integration matrix

Interface	**Producer Consumer**	**Input /output described**	**Storage**	**Contract documented?**
Web application to backend (all components)	React.js Node.js/Express.js	User inputs, images, and requests; results returned for display	MongoDB, as described in project documentation	No formal contract supplied
Wearable to platform (C1)	IoT device Node.js server	HR and SpO_2 values, checked on receipt	MongoDB, as described in project documentation	No formal contract supplied
Backend to worker-condition model (C1)	Node.js LightGBM model	HR and SpO_2 one of four worker-condition classes	To be confirmed	No formal contract supplied
Backend to disease model (C2)	Node.js Hugging Face API hosting YOLOv8s	Tea leaf image disease, confidence score, and treatment advice	MongoDB, as described in project documentation	No formal contract supplied
Backend to weather service (C3)	Node.js Python service	Region or weather data forecast, disease-risk estimate, and recommendations	To be confirmed	No formal contract supplied
Weather service to weather data source (C3)	Weather data source weather service	Weather records and readings	To be confirmed	No formal contract supplied
Backend to	Node.js →	Processed-tea	MongoDB/Mongo	No formal

grading model (C4)	Hugging Face API	image → predicted grade and confidence score	DB Atlas, as described in project documentation	contract supplied

Several components refer to MongoDB or MongoDB Atlas. However, these references do not establish that all components use the same database instance, collections, schema, or record identifiers. The available documentation also does not establish whether the components share user or estate identifiers, authentication mechanisms, or common timestamp conventions.

Before results from different components can be reliably correlated, the team should document the existing data structures and determine whether common identifiers and data conventions are required. Potential shared fields include estate identifiers, user identifiers, region identifiers, record identifiers, and timestamps. These should be treated as proposed standardization measures unless their use in the current implementation can be verified.

4.3 Shared non-functional requirements

This section describes the non-functional requirements relevant to the overall platform and the evidence available to assess them. No measured results for these requirements are documented in the supplied project materials.

Performance. The response times of image-based components depend on hosted model-inference services and network conditions. Performance targets and measured response times have not been documented.

Usability. The common dashboard should present the four decision-support areas clearly to estate managers and other users with limited technical experience. Formal usability evaluation results are not documented.

Security. Image uploads, worker records, and device data require appropriate authentication, access control, secure data transmission, and protection of external-service API keys. Although the grading component documentation refers to authentication and model services, a platform-wide security design has not been documented.

Privacy. Heart rate and SpO_2 readings may constitute sensitive personal data when associated with identifiable workers. Data collection should be limited to legitimate requirements, workers should receive appropriate information, and access should be restricted. Human review should be maintained when interpreting worker-condition predictions and deciding on subsequent actions.

Reliability. Dependence on hosted model-inference services and external weather-data sources may affect system availability. Appropriate handling of service failures, including error reporting and recovery behaviour, should be defined and evaluated, as discussed in Section 4.4.

Maintainability and scalability. The separation of the frontend, backend, and model-serving services can support independent updates and future expansion. However, model versioning, API versioning, and the procedures used to maintain compatibility between services are not documented in the available material.

Accessibility. Before deployment, the interface should be evaluated for usability under low-bandwidth conditions and for users with different levels of digital experience. Accessibility and connectivity-related evaluation results are not documented.

4.4 Integrated workflow and exception handling

During normal operation, the estate manager enters or uploads the required data, the backend forwards the request to the relevant component service, and the resulting output is stored and displayed through the web application, as described in Section 3.7. Table 4.2 summaries potential failure cases, the corresponding design considerations, and the implementation status documented in the available project materials.

According to the project paper, the tea leaf disease detection component checks whether an uploaded image contains a tea leaf, while the labor-management component generates alerts when HR or SpO_2 values exceed defined levels [1]. The implementation status of the other exception-handling behaviors listed in Table 4.2 is not documented in the available material. These behaviors are therefore presented as design considerations rather than confirmed system features.

Table 4.2: Failure cases, design considerations and status

Failure case	Design consideration
Model endpoint unavailable	Return a clear message to the user, retry with a limit, and avoid storing an empty record
Invalid image or image without a tea leaf	Reject before disease detection and ask the user for a new image
Missing weather records	Fall back to the most recent valid reading and flag the forecast as based on older data
Sensor transmission failure	Mark the worker's monitoring data as unavailable rather than inferring a condition; notify the supervisor
Database failure	Fail the request visibly, protect against partial writes and keep backups
Low-confidence prediction	Show the confidence score and advise review by an expert when below a defined threshold
HR or SpO_2 beyond defined levels	Alert the responsible person
Invalid user input	Validate on the client and on the server

4.5 Team contribution and dependency matrix

Table 4.3 summaries the four group members, their assigned research components, and the individual tasks described in the project documentation [1]. It also identifies dependencies on shared platform elements and external services. Shared responsibilities, including architecture design, dashboard consistency, integration activities, and preparation of the common report, should be confirmed against the team's actual contribution records. As the available documentation does not establish direct data dependencies between the four components, the dependencies listed primarily relate to the shared platform and the external services used by individual components.

Table 4.3: Team contribution and dependency matrix

Member	Component	Individual tasks described in [1]	Shared tasks	Dependencies
Gunasekara L.M.N.P (IT22267986)	1 Labor Management and Welfare Optimization	Worker and job module; wearable data handling; LightGBM model training; frontend and backend of the module	To be confirmed	Shared frontend shell, backend, database
Miuranga W.A.R (IT22209320)	2 Tea Leaf Disease Detection and Pattern Analysis	YOLOv8s model; image upload workflow; hosted inference; treatment advice; dashboard view	To be confirmed	Shared frontend shell, backend, database; Hugging Face API
Gunasekara G.N.D (IT22279620)	3 Tea-Specific Weather and Climate Intelligence	Weather data processing; prediction service and models; recommendations	To be confirmed	Shared frontend shell; weather data source
Pathirana I.M (IT22282668)	4 AI-Based Tea Sorting and Grading Automation	Grading web application; classification service integration; storage of results	To be confirmed	Shared frontend shell; hosted inference service; MongoDB

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

Table 5.1 separates the technologies that are common to the platform from those specific to a component. The common web layer consists of React.js, Node.js with Express.js and MongoDB. A single JavaScript-based web layer allows the four members to build module-specific views and routes on the same foundation. The component-specific technologies follow from the task each component performs: LightGBM for tabular physiological features, YOLOv8s [6] for object detection in images, a Python-based prediction service for weather forecasting, and a deep image classifier for grades. LightGBM [5] is a gradient boosting framework. The team's own reasons for each choice are given in the individual theses.

Table 5.1: Shared and component-specific technology stack

Layer	Technology	Used by
Frontend	React.js	All components
Backend	Node.js with Express.js	C1, C2, C4 stated; C3 via application backend
Database	MongoDB (MongoDB Atlas for C4)	C1, C2, C4; C3 to be confirmed
Hosted inference	Hugging Face API	C2, C4
Worker-condition model	LightGBM (LGBMClassifier); Google Colab for training; IoT wearable devices	C1
Disease model	YOLOv8s	C2
Weather service	Python Flask service; OpenWeatherMap API; model set to be confirmed	C3
Grading model	ConvNeXtV2-Tiny as reported; remote classification endpoint	C4

5.2 Development and integration process

Each component was developed by its assigned team member, while the machine learning models were trained separately according to the requirements of each component. For example, the LightGBM classifier was trained using Google Colab [1]. The developed components were subsequently brought together within the common web-based platform.

The available project documentation does not provide sufficient evidence to describe specific development practices, such as sprint planning, interface-first development, version-control conventions, code reviews, continuous integration and continuous deployment (CI/CD), issue tracking, or release management. Therefore, these practices are not presented as confirmed activities in this report.

To provide a more complete account of the development process, the team should verify and document the practices actually followed using available evidence, such as repository history, project plans, commit records, and task-tracking information.

5.3 Deployment and operational configuration

According to the project documentation, two components use model inference services hosted through the Hugging Face API. The documentation for the tea grading component also identifies MongoDB Atlas as a database service for storing records [1]. These details indicate the use of externally hosted services for selected parts of the system.

However, the available documentation does not sufficiently describe the hosting environments and deployment procedures for the frontend and backend. Details regarding environment-variable configuration, API-key management, application monitoring, database backups, data restoration, and service availability are also not documented. Therefore, a complete deployment topology cannot be established from the available evidence.

To provide a more comprehensive description of the deployment and operational configuration, the team should verify and document the hosting provider and environment used for each component, the deployment steps, configuration and

secret-management practices, the backup and restoration procedures, and any mechanisms used to monitor service availability and errors.

5.4 Evidence of team execution

Evidence of team execution can be demonstrated through project planning records, meeting minutes, version-control history, and individual research or development logs. However, these records were not available for review when this report was prepared. Consequently, this section does not make specific claims about team activities, meeting frequency, task completion, or development progress.

To strengthen the documentation of team execution, the team should prepare a concise summary of the available evidence, highlighting task allocation, individual contributions, coordination activities, and project progress where supported by records. The summary should be included in Appendix B, with appropriate references provided in this section. Any statements about team activities should be based on verifiable project records.

6 INTEGRATED TESTING, EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The evaluation of the proposed platform is organised into multiple layers to distinguish between model-level performance, component integration, end-to-end functionality, non-functional requirements, and user validation. These layers require different evaluation methods and supporting evidence.

The first layer focuses on the performance of each machine learning model using metrics appropriate to its specific task. For example, accuracy is used to report the performance of the four-class worker-condition classifier, while mean Average Precision at an Intersection over Union threshold of 0.5 (mAP@0.5), precision, recall, and F1-score are used to evaluate the tea leaf disease detector. The weather forecasting and tea grading components are evaluated using task-appropriate metrics, as reported in their respective component evaluations.

The second layer concerns interface and integration testing between the frontend, backend, model-serving services, and database. The third layer focuses on end-to-end testing to determine whether users can complete the intended workflows through the integrated platform. The fourth layer considers non-functional requirements, including security, performance, and usability. The fifth layer involves validation with representative users to assess whether the platform meets their practical needs.

Performance metrics from different components should not be directly compared or averaged to produce a single system accuracy. Each component addresses a different task and may use different datasets, class labels, and evaluation metrics. Therefore, the reported model-level results should be interpreted independently and should not be presented as evidence of overall platform accuracy.

The proposed acceptance criteria for the integrated platform should be agreed upon by the team and supported by documented test results. These criteria include successful completion of each of the four component workflows using valid inputs, appropriate user-visible responses when defined failure conditions occur, and the ability of representative users to complete relevant tasks through the dashboard.

These criteria are proposed evaluation targets and should not be interpreted as confirmed results unless supported by test evidence.

The available documentation does not establish a baseline comparison with existing manual practices or provide sufficient evidence of formal user validation. Such evidence would be required to assess the platform's practical benefits and determine whether it improves decision-making in real tea-estate environments.

6.2 Integration and end-to-end testing

Table 6.1 summarises the integration and end-to-end tests relevant to the proposed platform and identifies the outcomes supported by the available project evidence. The project paper describes the system as a web-based platform in which the four components were developed and evaluated using their respective datasets and trained models [1]. The components were also brought together within the running application.

However, no formal integration test logs, documented test cases, or recorded end-to-end test results were available for review when this report was prepared. Consequently, the available evidence is insufficient to mark individual integration or end-to-end tests as passed. The absence of documented test results does not necessarily mean that testing was not performed; rather, the outcomes cannot be independently verified from the supplied evidence.

To strengthen the evaluation, the team should document the test cases, input conditions, expected results, actual results, and test status for each component workflow. Any test marked as passed should be supported by recorded evidence, such as screenshots, execution logs, or completed test records.

Table 6.1: Integrated test plan and recorded outcomes

ID	Test	Scope
T1	Leaf image upload to displayed disease result	C2 workflow across frontend, backend, hosted model, database
T2	Wearable reading to displayed worker-condition class and alert	C1 workflow including device, backend, model, dashboard

ID	Test	Scope
T3	Region request to forecast and recommendations	C3 workflow
T4	Processed-tea image to displayed grade	C4 workflow
T5	Persistence and retrieval of results	All components and database
T6	Failure cases of Table 4.2	All
T7	Authentication and authorisation	Platform
T8	Response time under normal and peak load	Platform and hosted services
T9	Usability test with representative users	Dashboard
T10	User acceptance testing	Platform

6.3 Overall results

Table 6.2 summaries the component-level model performance results reported in the project paper and the group's component summaries [1]. Each result relates to a specific model and task and must be interpreted according to its dataset, evaluation method, and reported metric. Therefore, the results should not be directly compared or combined to calculate an overall system performance score.

The LightGBM-based worker-condition classifier achieved a reported accuracy of 82% across four worker-condition classes using the PhysioNet dataset. The tea leaf disease detector's reported mAP@0.5, precision, recall, and F1-score relate to its own evaluation dataset. These results describe the performance of the individual disease-detection model rather than the complete integrated platform.

The weather forecasting and tea grading results are also included as reported in the project documentation. However, the evaluation metric and methodology underlying the reported weather result of 89% require clarification. Similarly, the evaluation protocol and supporting evidence for the reported tea grading results of 99.7% require verification. These qualifications should be resolved and documented before the corresponding figures are presented as fully verified results.

Overall, the available results provide an indication of the reported performance of the individual machine learning components. However, they do not independently

establish the overall accuracy, operational effectiveness, or real-world suitability of the integrated platform. Further documented integration testing and, where feasible, validation under representative tea-estate conditions are required to assess these aspects.

Table 6.2: Consolidated component-level evaluation results

Component	Model approach	Dataset / test basis	Metric	Result	Qualification
C1	LightGBM classifier (LGBMClassifier)	PhysioNet Non-EEG Dataset for Assessment of Neurological Status	Accuracy	82%	Four classes; per-class results and split not stated here; public dataset, not estate workers
C2	YOLOv8s	Tea leaf disease image dataset; validation tests	mAP@0.5 Precision Recall F1-score	96.1% 95.2% 93.8% 94.5%	Applies to this model and dataset only; dataset size and classes in the component thesis
C3	Two configurations described (Table 3.2)	Weather records (details to be confirmed)	Overall accuracy	89% (as reported)	Model configuration and meaning of accuracy unconfirmed
C4	ConvNeXtV2-Tiny as reported	Processed-tea grade image dataset	Average precision Average recall Average F1-score	99.7% 99.7% 99.7%	Eight vs nine grades; training provenance and evaluation protocol unconfirmed

System-level testing is discussed separately in Section 6.2. Based on the available evidence, the conclusions that can be drawn at this stage remain limited. Each component has its own reported model-level evaluation results, and the project documentation describes the platform as a working web application [1]. However, the integrated behaviour of the complete system has not been demonstrated through documented integration and end-to-end test results. Therefore, the reported component-level results should not be interpreted as evidence of the overall performance or reliability of the integrated platform.

6.4 Stakeholder and user validation

Validation of the complete platform by representative stakeholders, including tea-estate managers, field supervisors, quality-control staff, and workers, is not documented in the available project evidence. No records of participant recruitment, user feedback, usability assessments, or stakeholder validation findings were available when this report was prepared. Therefore, no claims are made regarding the platform's acceptance, usability, or suitability for real-world tea-estate operations.

A structured validation process is recommended to evaluate the platform in a representative operational context. First, participants should be recruited from relevant stakeholder groups, with informed consent obtained before collecting feedback or personal data. Second, participants should be asked to complete scenario-based tasks covering the four component workflows. Third, evaluation measures should include task-completion rates, task-completion time, and responses to an appropriate standardised usability questionnaire.

In addition, qualified agricultural experts should review the disease-detection outputs and associated treatment recommendations, while tea-quality experts should assess the relevance of the predicted grading results. Worker-monitoring outputs should be reviewed with appropriate welfare, privacy, and data-protection safeguards to ensure that the results are interpreted responsibly and are not treated as definitive medical assessments.

The findings from this process should be documented and analysed to identify usability issues, operational limitations, and opportunities for improvement. Any resulting design changes should also be recorded. Until such validation is conducted and the results are documented, the practical suitability and stakeholder acceptance of the integrated platform remain unverified.

6.5 Achievement of common objectives

Table 6.3 maps the common project objectives to the available evidence and indicates the extent to which each objective is supported. The assessment distinguishes between component-level achievements, implementation within the

common platform, and objectives requiring further verification through documented system testing or real-world validation.

Objectives supported by reported model-level evaluation results are recognised as achieved at the component level, where applicable. Objectives relating to the development and presentation of the four components within the common web application are assessed using the available implementation evidence. However, objectives that require documented integration testing, formal usability assessment, or validation under actual tea-estate conditions are considered partially achieved or not yet verified, depending on the evidence available.

This assessment does not treat individual model metrics as proof of overall system performance. The status assigned to each objective should be supported by relevant project documentation, component evaluation results, or test records. Table 6.3 therefore presents an evidence-based assessment of objective achievement rather than assuming that all objectives have been fully achieved.

Table 6.3: Common objectives and supporting evidence

Objective	Evidence	Status	Justification
O1 Unified architecture	Architecture and workflow described [1]; Figures 3.1 and 3.2	Partly achieved	Architecture is described; running integration and shared contracts not documented
O2 Worker condition	Accuracy 82% [1]	Achieved at component level	Field validation and operational thresholds not documented
O3 Disease detection	mAP@0.5 96.1%, P 95.2%, R 93.8%, F1 94.5% [1]	Achieved at component level	Field validation and expert review of advice not documented
O4 Weather and risk	Accuracy 89% reported [1]	Partly achieved	Implemented configuration and metric definition unconfirmed
O5 Grading	Average P/R/F1 99.7% reported [1]	Partly achieved	Grade list and training provenance unconfirmed
O6 Evaluation and end-to-end test	Component metrics; no integration records	Partly achieved	End-to-end testing not documented

Objective	Evidence	Status	Justification
O7 Validation, security, privacy needs	Sections 6.4, 6.6, 7.4	Achieved as analysis	Needs are identified; work itself remains

6.6 Integrated limitations and threats

The main limitations arise from the gap between component-level evidence and system-level evidence. Datasets differ in origin and may not represent estate conditions: the worker-condition model was trained on a public dataset, and image models depend on the conditions under which images were collected. Reported metrics may not generalise to new estates, seasons, devices or camera conditions (threats to external validity). Differences between the project paper and the component documentation, particularly for the weather and grading components, threaten the internal consistency of the reported results. The dependency on externally hosted inference and weather services creates availability, cost and terms-of-use risks. Table 6.4 summarises the principal risks and the mitigation or future work for each.

Table 6.4: Risks, limitations and mitigation

Risk / limitation	Effect	Mitigation / future work
Metrics not validated in the field	Performance on real estates may be lower than reported	Pilot with real users and field data; report results by class
Inconsistent descriptions (weather model, grade list)	Unreliable report and unclear reproducibility	Confirm the implemented configuration; reconcile grade list; document datasets
Training provenance of grading model unverified	Training claims cannot be reproduced	Supply training scripts, dataset documentation and weights, or describe the component as integration of a hosted service
No documented interface contracts	Integration is fragile and hard to test	Write interface specifications and automated contract tests
Dependence on hosted services and connectivity	Features may fail when a service or network is unavailable	Define fallbacks and monitoring; evaluate offline options

Risk / limitation	Effect	Mitigation / future work
Worker health data privacy	Risk of misuse or loss of trust	Consent, minimisation, access control, human review of predictions
Over-reliance on automated advice	Wrong treatment or grading decisions	Show confidence; require expert review for low-confidence results

7 COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The main value proposition of the proposed platform is its potential to provide tea-estate managers with decision-support capabilities for four recurring operational areas through a single web-based application. These areas include workforce task allocation and worker-welfare monitoring, tea leaf disease identification and response guidance, weather forecasting and weather-related disease-risk assessment, and processed-tea grade classification.

By bringing these capabilities together within a common platform, the system could help managers access relevant information more conveniently when making operational decisions. Other potential beneficiaries include field supervisors responsible for coordinating work, workers whose welfare may be supported through appropriate monitoring and task recommendations, agricultural personnel who assess crop diseases, and quality-control staff involved in evaluating processed tea.

Table 7.1 identifies the principal stakeholder groups and their potential benefits from the proposed system. These benefits represent expected value propositions rather than measured outcomes. Improvements in productivity, worker welfare, disease management, weather-related decision-making, or grading consistency have not been established through documented field trials or formal stakeholder validation. Therefore, the practical benefits of the platform require further evaluation in representative tea-estate environments.

Table 7.1: Potential beneficiaries and value

Beneficiary	Potential value
Estate managers	Less time spent collecting information; more consistent, evidence-based decisions
Field workers	Possible earlier recognition of strain and fairer task distribution, if monitoring is introduced with consent and safeguards
Quality-control personnel	More consistent grading support and a record of graded batches

Beneficiary	Potential value
Estate owners and the wider industry	Possible improvements in productivity and quality consistency
Researchers and training institutions	A reference design and component-level results for further work

7.2 Market alternatives and differentiation

Potential alternatives to the proposed platform include existing manual practices, such as field observation and record keeping, as well as specialised tools that address individual tasks, including tea grading systems, plant disease classifiers, and weather-information services discussed in the project paper [1]. These alternatives provide different forms of support for specific operational needs.

According to the project paper, the tools considered generally focus on individual problem areas rather than bringing the four decision areas together within a common platform [1]. The proposed platform's potential differentiation therefore lies in combining workforce management and welfare monitoring, tea leaf disease detection, weather and disease-risk assessment, and processed-tea grading support within a single web-based application designed for tea-estate operations.

However, the available project evidence does not include a systematic competitor analysis, market survey, or customer interviews. Consequently, the size and structure of the target market, the availability of competing integrated solutions, customer demand, and the strength of the proposed differentiation have not been independently established. Further market research and stakeholder consultation would be necessary to assess the platform's competitive position and commercial potential.

7.3 Business model and feasibility

Several potential business models could be investigated for the proposed platform, including subscription-based access for individual tea estates, institutional licensing agreements with tea-estate companies, and installation and technical support services. The latter could include the supply and configuration of wearable devices where required. These models represent possible commercial approaches for future investigation rather than established business plans.

The main potential cost drivers include cloud hosting, externally hosted model-inference services, access to weather data, wearable hardware, technical support, and ongoing model maintenance. The actual costs would depend on the selected service providers, usage levels, deployment requirements, and the number of estates using the platform.

The working component prototypes described in the project paper provide preliminary evidence of technical feasibility [1]. However, operational feasibility and financial viability have not been formally assessed. No documented cost analysis, pricing study, revenue forecast, or field deployment assessment was available to establish whether the platform could be operated sustainably or adopted commercially. Further technical evaluation, stakeholder consultation, and cost-benefit analysis are therefore required before a viable commercial model can be recommended.

7.4 Adoption, intellectual property and compliance

Adoption barriers include limited connectivity in some estate areas, the cost of devices, the need for staff training, and the acceptance of health monitoring by workers and their representatives. Intellectual property and licensing need to be reviewed before commercial use: ownership of the code and models under SLIIT policy, the licence terms of the PhysioNet dataset and of any other training data, and the licence terms of the model frameworks and pretrained weights used (for example YOLOv8 [6] and ConvNeXt V2 [8]). Collection and processing of worker physiological data engage data-protection law, including the Personal Data Protection Act, No. 9 of 2022 of Sri Lanka [10], and the team should take legal advice before deployment. Table 7.2 summarises responsible-innovation considerations.

Table 7.2: Responsible-innovation considerations

Topic	Consideration
Privacy of worker physiological data	Informed consent, data minimisation, restricted access, retention limits; no use for punitive purposes
Human review of health-related predictions	Worker-condition classes are decision support, not diagnosis; a person must review before action

Topic	Consideration
Agricultural expert oversight	Treatment advice should be reviewed by an expert and not applied blindly
Model uncertainty	Show confidence scores and limits of each model; avoid presenting predictions as certain
Data security	Authenticate users, encrypt transport, protect API keys, back up data
Licensing of datasets and models	Check licences before commercial use
Access and reliability of connectivity	Design for intermittent networks and service failure

7.5 Sustainable Development Goal contribution

The platform could plausibly contribute to the goals listed in Table 1.2 by supporting worker welfare, more productive estate operations and better-informed responses to disease and weather. These contributions are potential, and no measurable SDG impact is claimed. Possible negative effects should also be considered: monitoring could be used to pressure workers, automation could change the tasks of experienced graders and inspectors, and reliance on connected technology could widen the gap between estates that can and cannot adopt it. Measures to sustain benefits and limit harm include consent and clear rules on the use of worker data, expert oversight, training for staff, affordable deployment options and periodic review of outcomes with the people affected.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

This project aimed to address the lack of a common decision-support platform for Sri Lankan tea-estate management across four operational areas: workforce welfare, tea leaf disease detection, weather-related risk assessment, and processed-tea quality classification. The available project evidence describes a web-based platform incorporating four components, each with its own machine learning model and component-level evaluation results.

The LightGBM-based worker-condition classifier reported an accuracy of 82% across four worker-condition classes. The tea leaf disease detector reported an mAP@0.5 of 96.1%, precision of 95.2%, recall of 93.8%, and an F1-score of 94.5%. The weather and tea grading components reported results of 89% and 99.7%, respectively, as described in the project documentation [1]. However, the evaluation metric associated with the weather result, the evaluation methodology and supporting evidence for the grading result, the grading class list, and aspects of the model configurations and training provenance require further verification before these figures can be treated as final results.

The project has brought the four component workflows together within a common web application. However, the available evidence does not fully establish the integrated behaviour of the platform through documented interface contracts, a verified common data schema, and formal end-to-end test results. Similarly, performance under real tea-estate conditions, usability among intended users, and operational aspects such as security, deployment reliability, and recovery procedures remain insufficiently verified.

Therefore, the project addresses the identified problem in part. The implemented components, their reported evaluation results, and their presentation within a common web application provide a foundation for an integrated decision-support solution. Nevertheless, further verification, system-level testing, and stakeholder validation are required before the platform's effectiveness in improving practical decision-making in Sri Lankan tea estates can be established.

8.2 Principal integrated contribution

The principal integrated contribution of this project is the presentation of four tea-estate decision-support components within a common web-based application. The platform brings together workforce welfare monitoring, tea leaf disease detection, weather-related risk assessment, and processed-tea quality classification, providing estate managers with a common access point for these distinct operational functions.

A further contribution is the consolidation of the reported component-level evaluation results obtained using different datasets, models, and evaluation methods. These results provide an overview of the individual components' reported performance, although they cannot be directly combined to establish the overall accuracy of the integrated platform.

The project also identifies the technical and operational requirements that need to be addressed to strengthen the integration of the individual components into a more cohesive system. These include documented interface contracts, a verified shared data model where appropriate, formal integration and end-to-end testing, stakeholder validation, and suitable privacy and security safeguards. Consequently, the contribution includes both the implemented common application and an evidence-based identification of the remaining requirements for a more thoroughly validated integrated decision-support platform.

8.3 Future directions

The following activities are recommended, in order of priority, to strengthen the evidence supporting the project and prepare the platform for potential deployment.

Verify and document component configurations: Confirm the implemented weather models and data sources, the tea grading model and its class list, and other relevant component configurations. Ensure consistency between the project paper and the individual theses.

Define and document interfaces: Specify the interface contracts and relevant data structures used for communication between the frontend, backend, model services, and database. Where a shared data schema is appropriate, define it clearly and introduce automated tests to verify interface compatibility.

Conduct systematic integration testing: Develop, execute, and document integration and end-to-end test cases for all four component workflows. Include the failure scenarios identified in Table 4.2 and record the expected and actual outcomes.

Perform stakeholder and expert validation: Evaluate the platform with representative tea-estate managers, field supervisors, quality-control staff, and workers. Seek appropriate expert review of disease-treatment recommendations and processed-tea grading outputs.

Strengthen datasets and evaluation methods: Expand and improve dataset quality, report per-class performance where appropriate, verify the separation of training and testing data, and evaluate model performance using representative field data.

Improve security, privacy, and compliance: Establish suitable safeguards for worker physiological data, review access controls and data-protection practices, and verify applicable legal obligations and dataset, framework, and model licences before deployment or commercial use.

Establish operational readiness: Document deployment procedures, configuration management, monitoring, backup, and restoration. Test the platform's behaviour when external model services, weather-data services, or network connections become unavailable.

Explore further system capabilities: Investigate potential cross-component decision-support functions, such as using weather-related disease-risk information to prioritise leaf inspections or using worker-condition information to inform task allocation. Future work could also consider additional IoT devices and a mobile application, subject to feasibility assessment and appropriate validation [1].

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APPENDICES

Appendix A Integrated interface specification

Table A.1 summaries the interface information identified in the project documentation and distinguishes documented details from implementation information that still requires verification.

Table A.1: Interface specification status

ID	Description (from [1])	Route / method	Request and response format	Authentication, validation, errors
I1	Frontend to backend requests for all four modules	Not documented	Not documented	Not documented
I2	Wearable HR and SpO_2 values to Node.js server	Not documented	Not documented	Values are checked on receipt [1]; rules not documented
I3	Backend to worker-condition model	Not documented	HR, SpO_2 in; one of four classes out	Not documented
I4	Backend to Hugging Face API (YOLOv8s)	Not documented	Image in; disease, confidence, advice out [1]	Leaf-presence check [1]; others not documented
I5	Backend to weather service	Not documented	Region / reading in; forecast, risk, recommendations out [1]	Not documented
I6	Weather service to data source	Not documented	Not documented	Not documented
I7	Backend to Hugging Face API (grading)	Not documented	Image in; grade and confidence out [1]	Not documented

A.2 System User Interface Screenshots

This section presents selected screenshots of the implemented TeaGuard web application to illustrate the user interface and the workflows available through the common platform. Each figure should represent an actual screen captured from the running application. The screenshots provide visual evidence of the implemented

interface and should not be interpreted as proof of formal integration testing or user validation.

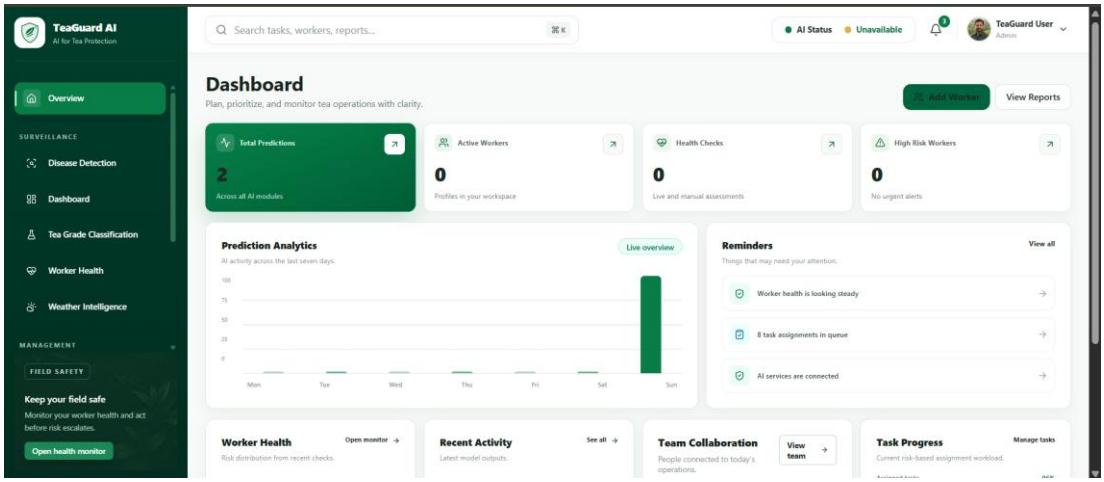


Figure A.1: Main Dashboard of the TeaGuard Platform

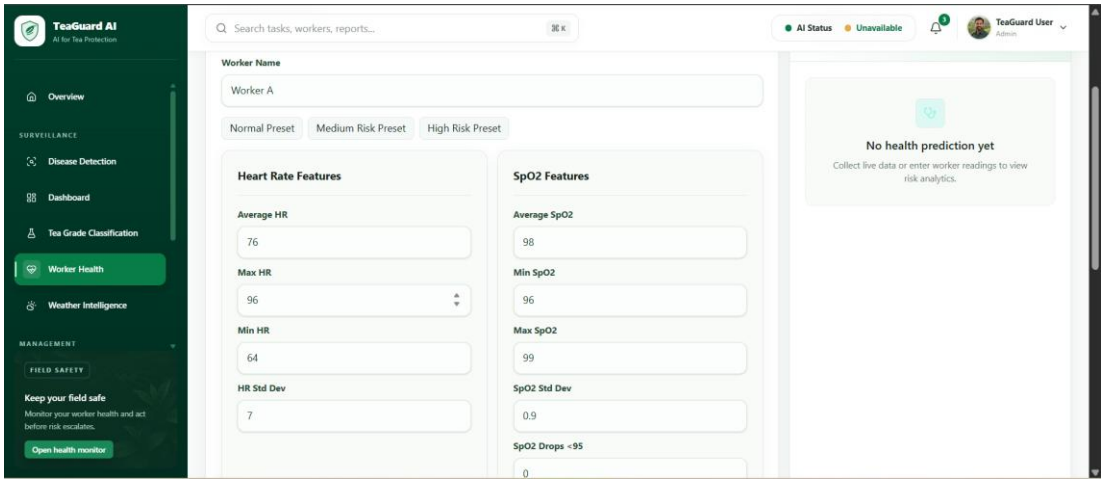


Figure A.2: Labor Management and Worker-Condition Monitoring Interface

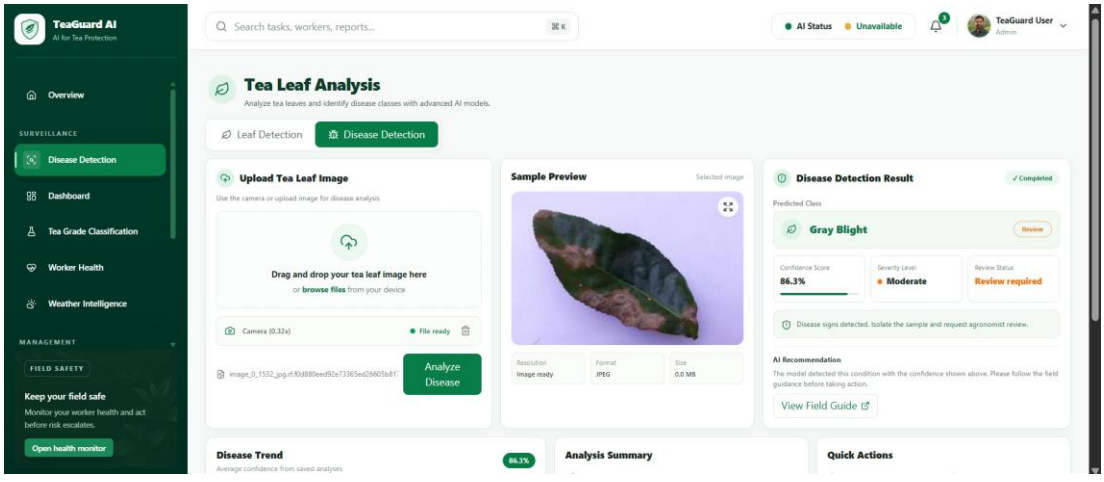


Figure A.3: Tea Leaf Disease Detection Interface

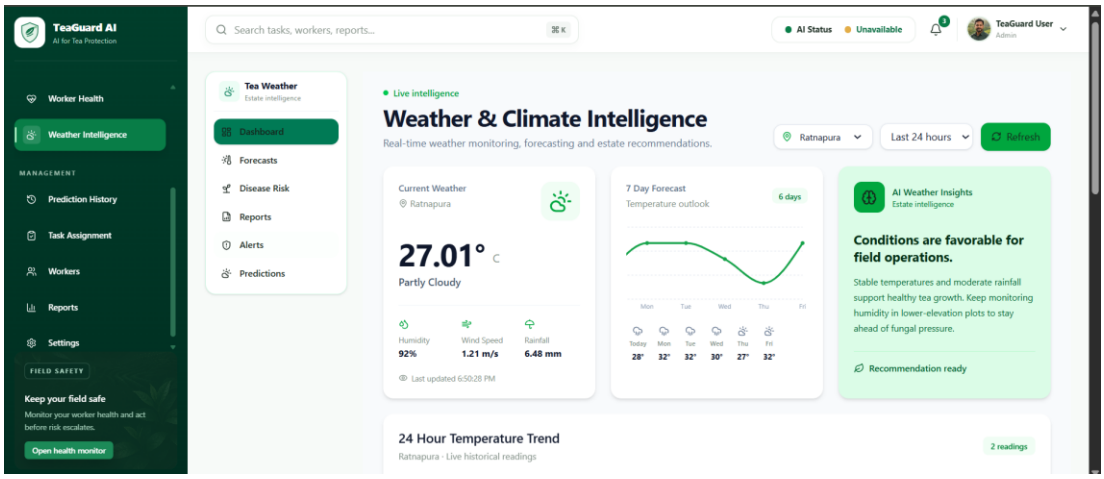


Figure A.4: Weather and Climate Intelligence Interface

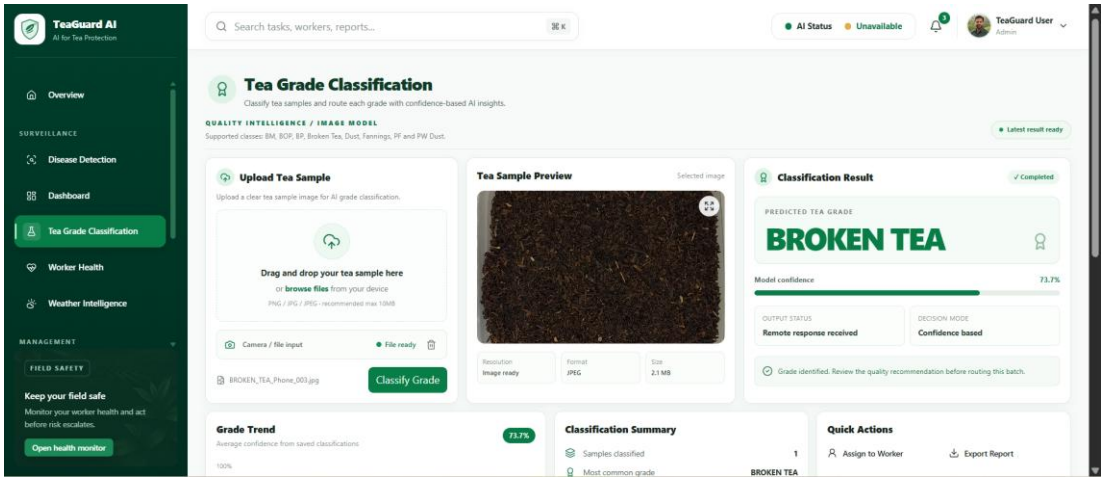


Figure A.5: Tea Sorting and Grading Interface

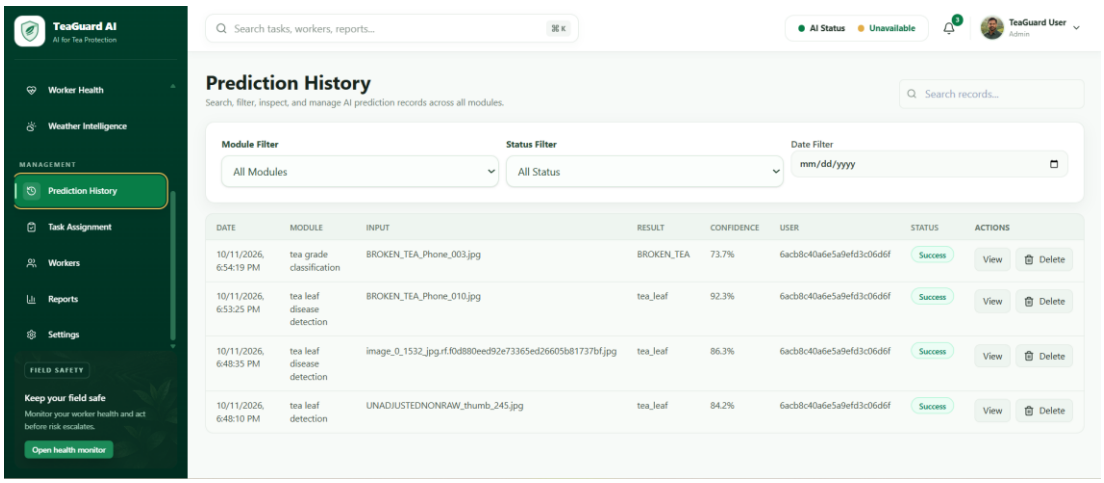


Figure A.6: Prediction Results and Decision-Support Outputs

Appendix B Team contribution declaration

Table B.1 presents the individual contributions of the four group members to the project, based on the project documentation [1]. Shared contributions and individual confirmations must be reviewed, agreed upon, and completed by all four members before submission.

Table B.1: Team contribution declaration

Member	Component	Individual Contribution	Shared Contribution
Miuranga W.A.R (IT22209320)	2. Tea Leaf Disease Detection and Pattern Analysis	YOLOv8s-based disease detection, image-processing workflow, and treatment recommendations.	To be confirmed and completed jointly by the team.
Gunasekara L.M.N.P (IT22267986)	1. Labor Management and Welfare Optimization	Worker and job management module, wearable data handling, and LightGBM classifier.	To be confirmed and completed jointly by the team.
Gunasekara G.N.D (IT22279620)	3. Tea-Specific Weather and Climate Intelligence	Weather forecasting and disease-risk assessment service, including recommendations.	To be confirmed and completed jointly by the team.
Pathirana I.M (IT22282668)	4. AI-Based Tea Sorting and Grading Automation	Tea grading web application and classification-service integration.	To be confirmed and completed jointly by the team.